

Understanding Uber Ride Cancellations:

A Data-Driven Study

STA 4365

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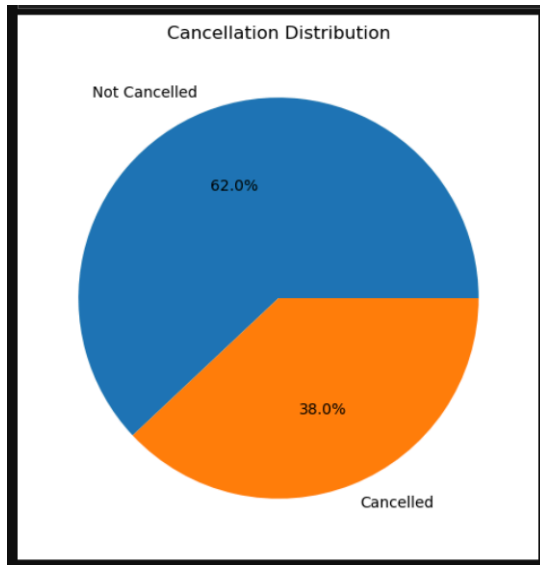
Introduction

Ride cancellations are a significant issue in ride-sharing platforms like Uber because they lead to wasted time, higher operational costs, and a negative experience for both riders and drivers. When cancellations happen frequently, they affect the reliability of the service and make it harder for the platform to efficiently match riders with available drivers. This not only impacts customer satisfaction but also reduces overall system efficiency. Therefore, understanding the reasons behind ride cancellations is essential in order to improve service quality and optimize platform performance.

The Kaggle dataset (link is at the end of the report) provides detailed information on Uber ride bookings in 2024, enabling analysis of booking behavior, cancellation patterns, and operational performances. It includes approximately 150,000 total bookings across multiple vehicle types, with features such as pickup and drop locations, ride timing, customer and driver behavior, and ride outcomes. This dataset allows for both exploratory analysis and predictive modeling of ride cancellations, which is a key operational challenge in ride-sharing platforms.

In on-demand platforms (ride-sharing, delivery, logistics), cancellations create inefficiencies, increase costs, and negatively impact user satisfaction. To build a predictive model for ride cancellations and analyze the most important factors affecting customer and driver cancellation behavior.

Target Variable



In our project, we focused on identifying and analyzing the target variable, which represents the main outcome we are trying to predict. The target variable is important because it guides the entire modeling process and helps us evaluate how well our model performs. However, the dataset did not originally contain a direct target variable for cancellation prediction. Instead, it included a column called “*Booking Status*” with multiple categories such as *Completed*, *Cancelled by Driver*, *Cancelled by Customer*, *No Driver Found*, and *Incomplete*.

To address this, we created a new target variable called *is_canceled*. We converted the multi-class booking status into a binary format, where all types of unsuccessful or canceled rides (Cancelled by Driver, Cancelled by Customer, No Driver Found, and Incomplete) were labeled as 1, and successful rides (Completed) were labeled as 0. This transformation allowed us to simplify the problem into a classification task, making it easier to apply predictive models and analyze the factors that contribute to ride cancellations.

Data Preprocessing

Before building any models, we performed several data preprocessing steps to clean and prepare the dataset for analysis. These steps were required to ensure data quality, consistency, and suitability for modeling.

Understanding the Dataset Structure

We first used basic functions to explore the dataset, including checking the number of rows, columns, data types, and non-null values. The dataset contains 150,000 entries and 21 columns, including both numerical and categorical features such as booking details, ride information, and customer/driver behavior.

	Avg VTAT	Avg CTAT	Cancelled Rides by Customer	Cancelled Rides by Driver	Incomplete Rides	Booking Value	Ride Distance	Driver Ratings	Customer Rating
count	139500.000000	102000.000000	10500.0	27000.0	9000.0	102000.000000	102000.000000	93000.000000	93000.000000
mean	8.456352	29.149636	1.0	1.0	1.0	508.295912	24.637012	4.230992	4.404584
std	3.773564	8.902577	0.0	0.0	0.0	395.805774	14.002138	0.436871	0.437819
min	2.000000	10.000000	1.0	1.0	1.0	50.000000	1.000000	3.000000	3.000000
25%	5.300000	21.600000	1.0	1.0	1.0	234.000000	12.460000	4.100000	4.200000
50%	8.300000	28.800000	1.0	1.0	1.0	414.000000	23.720000	4.300000	4.500000
75%	11.300000	36.800000	1.0	1.0	1.0	689.000000	36.820000	4.600000	4.800000
max	20.000000	45.000000	1.0	1.0	1.0	4277.000000	50.000000	5.000000	5.000000

We also looked at the numerical features to get a better idea of how the data is spread out. For example, the average booking value is around 508, but it ranges a lot—from as low as 50 to over 4000—showing that ride prices can vary quite a bit. Ride distance has an average of about 24.6, with values going from 1 to 50, which means trips can be very short or fairly long. Driver and customer ratings are mostly high, usually above 4, which suggests that most users have a good experience. We also noticed that features like Avg VTAT and Avg CTAT vary quite a bit, which could play a role in whether a ride gets completed or canceled.

Data Type Conversion

Some columns were not in the appropriate format for analysis. For example, the Time column was originally stored as an object (string). We converted it into a proper datetime format to allow time-based analysis and feature extraction if needed.

Handling Missing Values

We analyzed missing values across all columns using `.isnull()` functions. Several columns had a high percentage of missing data.

Incomplete Rides Reason	141000	Incomplete Rides Reason	94.0
Incomplete Rides	141000	Incomplete Rides	94.0
Cancelled Rides by Customer	139500	Cancelled Rides by Customer	93.0
Reason for cancelling by Customer	139500	Reason for cancelling by Customer	93.0
Driver Cancellation Reason	123000	Driver Cancellation Reason	82.0
Cancelled Rides by Driver	123000	Cancelled Rides by Driver	82.0
Customer Rating	57000	Customer Rating	38.0
Driver Ratings	57000	Driver Ratings	38.0
Ride Distance	48000	Ride Distance	32.0
Booking Value	48000	Booking Value	32.0
Payment Method	48000	Payment Method	32.0
Avg CTAT	48000	Avg CTAT	32.0
Avg VTAT	10500	Avg VTAT	7.0
Time	0	Time	0.0
Drop Location	0	Drop Location	0.0
Pickup Location	0	Pickup Location	0.0
Vehicle Type	0	Vehicle Type	0.0
Customer ID	0	Customer ID	0.0
Booking Status	0	Booking Status	0.0
Booking ID	0	Booking ID	0.0
Date	0	Date	0.0
dtype: int64		dtype: float64	

For example:

- Incomplete Rides and Incomplete Rides Reason had about 94% missing values
- Cancelled Rides by Customer and related reason columns had around 93% missing values
- Driver-related cancellation columns had about 82% missing values

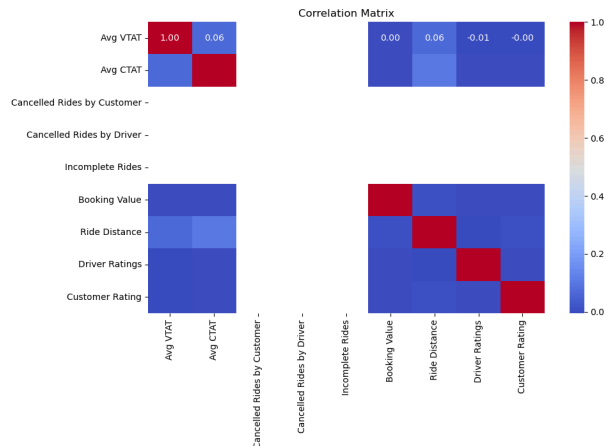
- Other features like ratings, ride distance, and booking value had moderate missing values (30–40%)

These missing values make sense because some columns are only filled in certain situations. For example, cancellation reasons are only recorded when a ride is actually canceled, so it's normal to see a lot of missing values in those columns. While handling missing data, we used different approaches based on the type of feature. For numerical columns like ride distance, booking value, and ratings, we filled missing values using the median since it is less affected by extreme values. For categorical columns, such as payment method, we replaced missing values with the most common category (mode). This helped us keep the data consistent without distorting the overall patterns in the dataset.

Understanding Special Columns

Some columns, such as Incomplete Rides, Cancelled Rides by Driver, and Cancelled Rides by Customer, contained only two types of values: 1 indicating the event occurred and NaN indicating no event. This helped us interpret these features as indicator variables (binary-like behavior), which is useful during modeling.

Checking Feature Relationships



We also used a correlation matrix to understand the relationships between numerical features. Most variables showed very weak or no correlation (values close to 0), meaning they provide independent information. A few features, like ride distance and Avg CTAT, had slight positive correlation, but it was still weak.

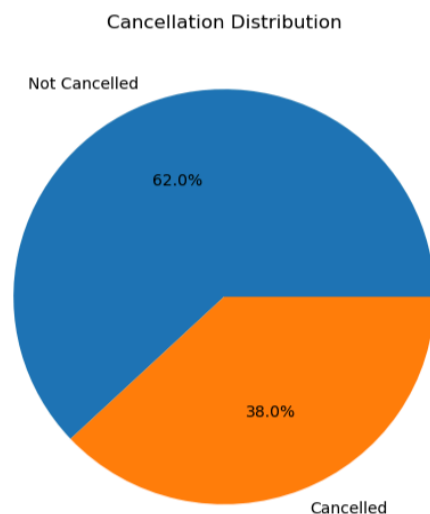
Some areas in the heatmap were blank due to high missing values in columns related to cancellations and incomplete rides, making it difficult to compute correlations. Overall, correlation was not very helpful for handling missing values, so we will use more practical methods like median and mode instead.

Exploratory Data Analysis

The goal of EDA is to identify patterns, detect anomalies, and gain insights into how different features behave and relate to each other. In this project, we used various visualizations to better understand the distribution of variables, relationships between features, and key factors that may influence ride cancellations. This step helped us make better decisions for data preprocessing and model building. We performed EDA based on Location, Time, Payment and Vehicle Type.

Understanding the Target Variable (is_cancelled)

To better understand our target variable, we looked at the distribution of the newly created *is_cancelled* column.. This column was created by converting different booking statuses into a binary format, where 1 represents cancelled or unsuccessful rides and 0 represents completed rides.



From the results, we can see that about 62% of the rides were completed (0), while around 38% were cancelled or not completed (1). This shows that cancellations make up a significant portion of the dataset, but the data is still fairly balanced. Understanding this distribution is important because it helps us decide how to train our model and whether we need to handle any imbalance in the data.

Analysis of Cancellation Reasons

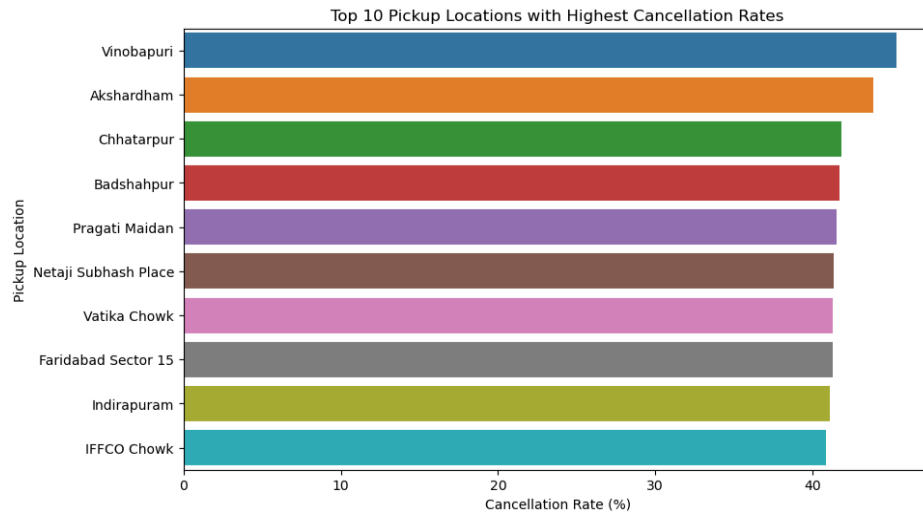
We looked at the most common reasons for rides being cancelled by drivers. The top reasons include customer-related issues, customers being sick, personal or car-related issues, and situations where more than the permitted number of people were present.

Driver Cancellation Reason	
Customer related issue	6837
The customer was coughing/sick	6751
Personal & Car related issues	6726
More than permitted people in there	6686

These factors could directly affect the target variable (*is_cancelled*), since they increase the likelihood of a ride being cancelled. For example, if a driver faces vehicle issues or a customer violates ride rules, the ride is more likely to be cancelled.

From a business perspective, Uber can reduce cancellations by improving communication with customers (e.g., clear ride policies), providing better support for drivers (like maintenance assistance), and adding stricter checks for rule violations. This can help improve ride completion rates and overall user experience.

Cancellation Analysis by Pickup Location



Pickup Location	Total Rides	Count	Rate (%)
Vinobapuri	823	373	45.321993
Akshardham	839	368	43.861740
Chhatarpur	829	347	41.857660
Badshahpur	868	362	41.705069
Pragati Maidan	920	382	41.521739
Netaji Subhash Place	830	343	41.325301
Vatika Chowk	833	344	41.296519
Faridabad Sector 15	831	343	41.275572
Indirapuram	854	351	41.100703
IFFCO Chowk	830	339	40.843373

We analyzed cancellations based on pickup locations to see if certain areas have higher cancellation rates. From the results, locations like Vinobapuri, Akshardham, and Chhatarpur show the **highest cancellation rates, all around 40–45%**. This means that nearly half of the rides in these areas end up being cancelled.

In addition to that, we used three metrics to better understand cancellations across different locations. *Total rides* is calculated using `.size()`, which gives the total number of bookings for each pickup location. *Count* is calculated using the sum of the `is_cancelled` column, which tells us how many of those rides were actually cancelled. The *cancellation rate* is the percentage of cancelled rides out of total rides.

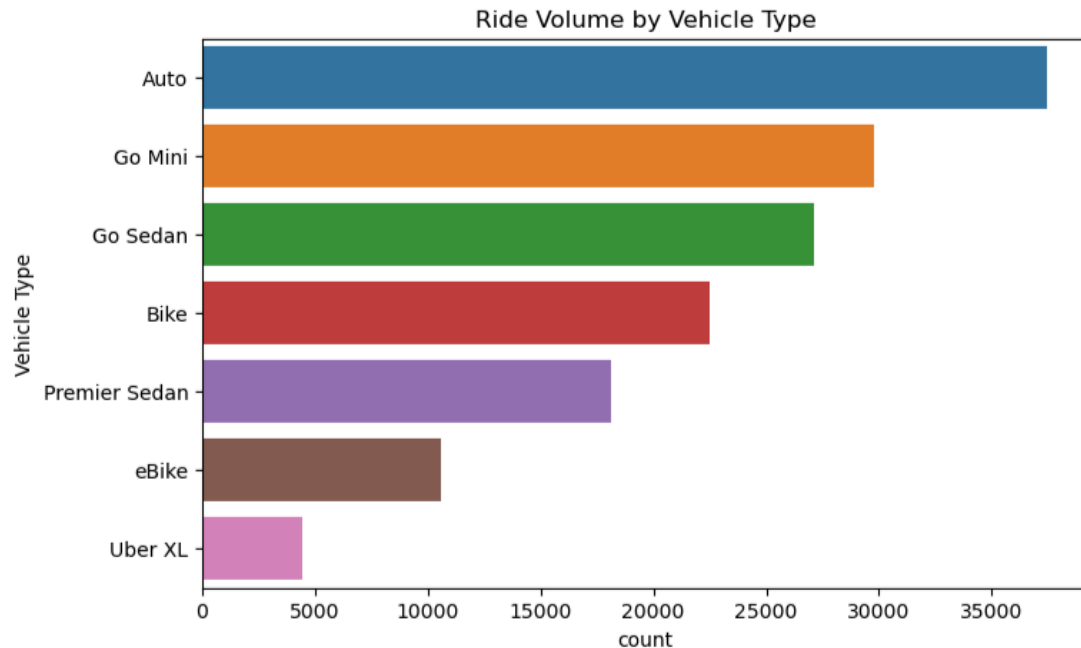
These three metrics together help us understand the situation more clearly. Total rides show how busy a location is, count shows how many cancellations are happening, and rate shows how serious the cancellation problem is relative to the number of rides. For example, a location with many rides may have a high count of cancellations just because of high demand, but its rate might still be moderate. On the other hand, a location with fewer rides but a high rate indicates a bigger issue that needs attention.

The bar chart clearly shows which pickup locations have the highest cancellation rates, making it easier to compare across areas. These patterns directly affect the target variable (*is_cancelled*), since location becomes an important factor in predicting whether a ride will be cancelled. Possible reasons for this could include **low driver availability, traffic congestion, or inefficient matching** between drivers and riders in certain areas.

To reduce cancellations, Uber can focus on these high-risk locations by improving driver availability, offering incentives in busy areas, and optimizing ride matching. This can help lower cancellation rates and improve overall service.

Cancellation Analysis by Vehicle Type

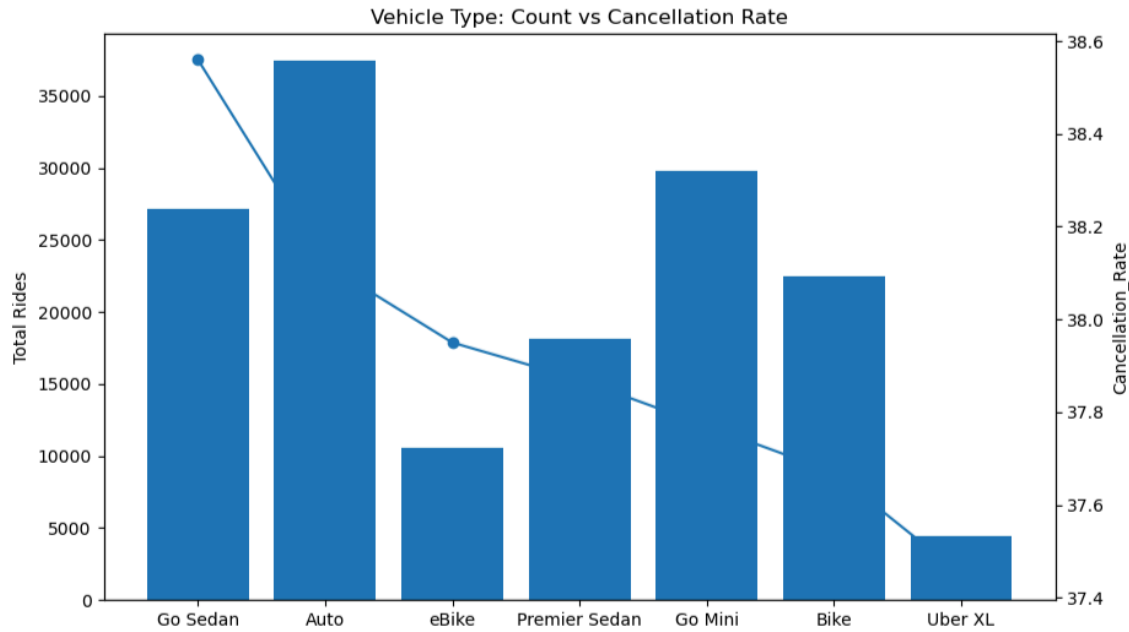
Ride Volume by Vehicle Type



We analyzed the distribution of rides across different vehicle types to understand demand patterns. The results show that **Auto, Go Mini, and Go Sedan** have the highest number of bookings, indicating that they are the most preferred options among users. In contrast, **Uber XL and eBike** have significantly lower ride volumes, suggesting limited demand for these categories.

From a business perspective, this indicates that Uber's core demand is concentrated in affordable and mid-range vehicle options. To improve performance, Uber can focus on increasing driver availability and optimizing operations for high-demand categories like Auto and Go Mini. For lower-demand categories such as Uber XL, targeted promotions or pricing strategies could help increase usage.

Cancellation Rate by Vehicle Type



We also compared total rides with cancellation rates across vehicle types. The analysis shows that cancellation rates are relatively consistent across all categories, generally ranging between **37% and 38%**. However, **Go Sedan** has a slightly higher cancellation rate, while **Uber XL** shows a comparatively lower rate.

This suggests that vehicle type alone does not strongly influence cancellations, and other factors such as location, timing, or driver behavior may have a greater impact. From a business standpoint, Uber can focus on improving driver allocation and matching efficiency across all vehicle types rather than targeting a specific category. Additionally, slight improvements in categories with higher cancellation rates, like Go Sedan, could help reduce overall cancellations and improve customer satisfaction.

Time-Based Analysis of Cancellations

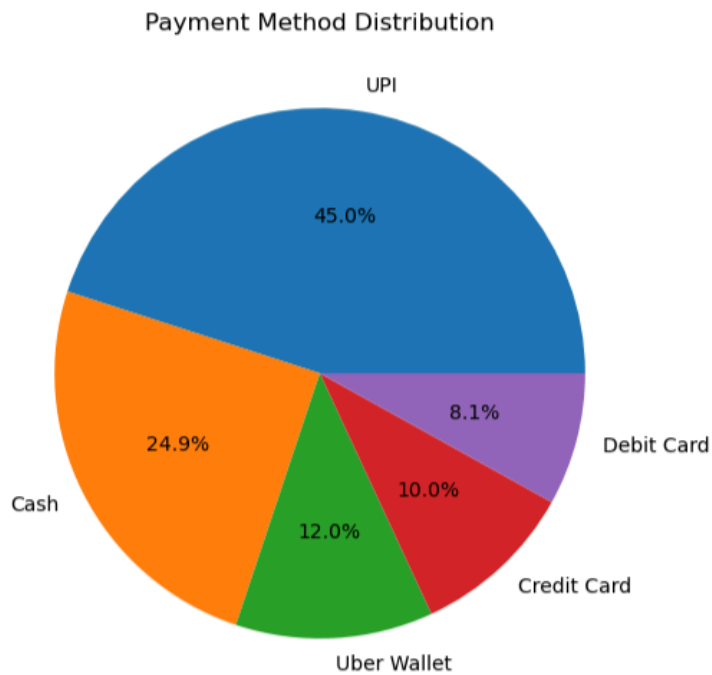
day_name		time_period	
Monday	38.467936	Afternoon	38.270580
Tuesday	38.427376	Early Morning	38.012151
Wednesday	37.906879	Evening	38.165828
Thursday	37.751591	Morning	37.708484
Friday	38.061410	Night	37.566069
Saturday	37.600966		
Sunday	37.779232		

We analyzed how cancellations vary across different hours of the day and grouped them into broader time periods such as early morning, morning, afternoon, evening, and night. The results show that cancellation rates remain fairly consistent throughout the day, generally ranging between **37% and 38%**. Slightly higher cancellation rates are observed during early morning and afternoon hours.

We also examined cancellation patterns across days of the week. The results indicate that cancellation rates are very similar across all days, with only small variations. For example, Monday and Tuesday show slightly higher cancellation rates, while Thursday and Saturday are slightly lower. However, the overall difference across days is minimal.

From a business perspective, this suggests that **time of day and day of week are not strong factors affecting cancellations**. Instead, cancellations seem to be influenced more by other factors such as location, driver availability, or operational issues. To improve performance, Uber can focus less on time-based strategies and more on optimizing driver supply, improving matching efficiency, and addressing location-specific challenges. However, slight adjustments during early morning or peak hours (such as increasing driver incentives) may still help reduce cancellations marginally.

Payment Method Distribution



We analyzed the distribution of payment methods used for bookings. The pie chart shows that **UPI is the most commonly used payment method (45%)**, followed by **Cash (24.9%)**. Other methods like **Uber Wallet (12%)**, **Credit Card (10%)**, and **Debit Card (8.1%)** are used less frequently.

From a business perspective, this suggests that payment method is not a strong factor influencing cancellations. Therefore, Uber should focus more on other factors like location, driver availability, or ride conditions rather than payment type when trying to reduce cancellations.

Cancellation Rate by Payment Method

Payment Method	
Cash	0.088816
Credit Card	0.087080
Debit Card	0.086540
UPI	0.088763
Uber Wallet	0.087162

While the previous analysis showed the distribution of different payment methods, we further examined whether payment type has any impact on cancellation rates. The results show that cancellation rates are very similar for all methods, with only slight differences. This means that whether a user pays using UPI, cash, card, or wallet does not significantly affect the likelihood of a ride being cancelled.

From a business perspective, this suggests that payment method is not a strong factor influencing cancellations. Therefore, Uber should focus more on other factors like location, driver availability, or ride conditions rather than payment type when trying to reduce cancellations.

Numeric Feature Comparison

	Booking Value	Ride Distance	Avg CTAT	Avg VTAT	Driver Ratings	Customer Rating
is_cancelled						
0	508.178215	26.000493	30.034772	8.511878	4.230992	4.404584
1	509.512111	10.547706	20.003233	8.345299	NaN	NaN

We compared the average values of key numerical features for completed rides (*is_cancelled* = 0) and cancelled rides (*is_cancelled* = 1) to understand how they differ.

From the results, we observed that **ride distance is much lower for cancelled rides**, which suggests that shorter trips are more likely to be cancelled. Similarly, **average CTAT (trip duration)** is lower for cancelled rides, indicating that shorter trips tend to have higher cancellation chances. On the other hand, **booking value and Avg VTAT (driver arrival time)** are almost the same for both cancelled and completed rides, meaning they are not strong factors in distinguishing cancellations.

We also noticed that **driver and customer ratings are missing for cancelled rides**, since ratings are only given after a ride is completed. Therefore, these features are not useful for predicting cancellations.

	Avg CTAT	Avg VTAT	customer_cancelled	driver_cancelled
Avg CTAT	1.000000	0.062175	NaN	NaN
Avg VTAT	0.062175	1.000000	0.306655	-0.123904
customer_cancelled	NaN	0.306655	1.000000	-0.128540
driver_cancelled	NaN	-0.123904	-0.128540	1.000000

To further understand cancellations, we created two new features: *customer_cancelled* and *driver_cancelled*. We then analyzed their relationship with time-based features like Avg VTAT and Avg CTAT. The results show that **customer cancellations are associated with higher VTAT (waiting time)**. This means that when drivers take longer to reach the pickup location, customers are more likely to cancel the ride. CTAT could not be analyzed properly due to missing values in cancelled rides.

Overall, the results suggest that waiting time plays a major role in ride cancellations, especially from the customer side. Uber can reduce cancellations by improving driver allocation, reducing pickup time, and ensuring better matching between drivers and riders. Additionally, focusing on short-distance rides and improving efficiency in those cases could further help reduce cancellation rates.

Treating Missing Values

Handling missing values was a critical preprocessing step to ensure the dataset remained reliable and suitable for modeling. Columns with more than **90% missing values** were removed entirely, since such high levels of missingness make the data unreliable and add little meaningful information to the model. For the remaining features, appropriate imputation techniques were applied based on the type of data. Numerical variables such as **Booking Value**, **Ride Distance**, **Average CTAT**, and **Average VTAT** were imputed using the **median**, which is preferred because it is less sensitive to extreme values and outliers. For categorical variables like **Payment Method**, missing values were filled using the **mode**, ensuring that the most frequently occurring category was used to maintain realistic data distribution.

This approach ensured that no rows had to be dropped unnecessarily, preserving the overall dataset size while maintaining data integrity. After these steps, the dataset contained **no**

```
df.isnull().mean().sort_values(ascending=False)
```

Incomplete Rides	0.94
Incomplete Rides Reason	0.94
Reason for cancelling by Customer	0.93
Cancelled Rides by Customer	0.93
Driver Cancellation Reason	0.82
Cancelled Rides by Driver	0.82
Customer Rating	0.38
Driver Ratings	0.38
Payment Method	0.32
Booking Value	0.32
Ride Distance	0.32
Avg CTAT	0.32
Avg VTAT	0.07
is_cancelled	0.00
hour	0.00
time_period	0.00
day_of_week	0.00
day_name	0.00
customer_cancelled	0.00
Date	0.00
Time	0.00
Drop Location	0.00
Pickup Location	0.00
Vehicle Type	0.00
Customer ID	0.00
Booking Status	0.00
Booking ID	0.00
driver_cancelled	0.00
dtype: float64	

remaining missing values, allowing the model to train effectively without errors or bias introduced by null entries. Proper handling of missing data contributed significantly to the stability and performance of the final model.

Model Building

After completing data preprocessing and EDA, we prepared the dataset for model building. First, we removed unnecessary columns such as Time, Date, and day_name since they were either redundant or already captured in other features like hour and day of week. Next, we converted all categorical variables into numerical format using one-hot encoding (get_dummies).

After encoding, we checked for missing values and confirmed that there were no null values left in the dataset. This ensured that the data was clean and ready for modeling.

Finally, we separated the dataset into input features (X) and the target variable (y). The is_cancelled column was used as the target variable, and it was removed from X to avoid data leakage. This prepared our data for training machine learning models to predict ride cancellations.

Train-Test Split

```
: from sklearn.naive_bayes import GaussianNB
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import SGDClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.ensemble import GradientBoostingClassifier

: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

We then split the dataset into training and testing sets using an 80-20 split. This means 80% of the data was used to train the model, and 20% was used to evaluate its performance. A random state was set to ensure consistent results.

We also imported multiple machine learning algorithms, including Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, Naive Bayes, Gradient Boosting, and XGBoost. These models were used to compare performance and identify the best approach for predicting ride cancellations.

Model Selection:

```
models = []
models.append(('Logistic Regression', LogisticRegression(random_state=42)))
models.append(('Naive bayes', GaussianNB()))
models.append(('KNN', KNeighborsClassifier()))
models.append(('Random Forest', RandomForestClassifier(random_state = 42)))
models.append(('SGD Classifier', SGDClassifier(random_state=42)))
models.append(('XGBoost', XGBClassifier(random_state = 42)))
models.append(('GradientBoosting', GradientBoostingClassifier(random_state = 42)))
results = []
names = []

for name, model in models:
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    print("{} : {}".format(name, accuracy))
```

C:\Users\tsrap\anaconda3\Lib\site-packages\sklearn\linear_model_logistic.py:458: Con
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
Logistic Regression : 0.8102333333333334
Naive bayes : 0.7041
KNN : 0.9439
Random Forest : 0.9579
SGD Classifier : 0.6746333333333333
XGBoost : 0.9558
GradientBoosting : 0.9579333333333333

We trained multiple machine learning models to predict ride cancellations, including Logistic Regression, Naive Bayes, K-Nearest Neighbors (KNN), Random Forest, and XGBoost. Each model was trained using the training dataset and then evaluated on the test dataset using accuracy as the performance metric.

From the results, we observed that Random Forest and Gradient Boosting performed the best, both achieving accuracy close to 95–96%, followed closely by XGBoost. KNN also performed well with around 94% accuracy, while Logistic Regression and Naive Bayes had lower performance in comparison.

Overall, tree-based models like Random Forest and XGBoost performed better because they can handle complex patterns and interactions in the data more effectively.

Models:

XGBoosting:

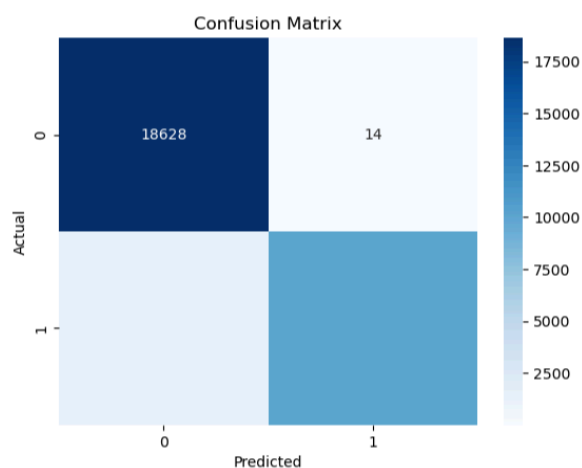
XGBoost is one of the best-performing models (after random forest) to predict ride cancellations. The goal was to build a model that can accurately classify whether a ride will be cancelled or not, while also understanding which factors are most important.

The model achieved high performance, with training and testing accuracy both around 95.8%, and the cross-validation score was also very similar (~95.7%). The difference between training and testing accuracy is very small, which means the model is well-balanced and not overfitting or underfitting. This shows that the model generalizes well to new data.

```
: cv_acc = xgb_cv_scores.mean()

print(f"Training Accuracy: {train_acc:.4f}")
print(f"Cross-Validation Accuracy: {cv_acc:.4f}")
print(f"Test Accuracy: {test_acc:.4f}")

Training Accuracy: 0.9575
Cross-Validation Accuracy: 0.9567
Test Accuracy: 0.9580
```



```
importance = pd.DataFrame({
    'Feature': X_train.columns,
    'Importance': xgb.feature_importances_
}).sort_values(by='Importance', ascending=False)

print(importance.head(10))
```

	Feature	Importance
3	Ride Distance	0.617486
1	Avg CTAT	0.135123
0	Avg VTAT	0.044002
364	Payment Method_UPI	0.005323
2	Booking Value	0.003623
97	Pickup Location_Lok Kalyan Marg	0.003358
233	Drop Location_Gurgaon Sector 56	0.003243
132	Pickup Location_Patel Chowk	0.003130
191	Drop Location_Anand Vihar ISBT	0.003039
351	Drop Location_Udyog Vihar	0.003013

To better understand model performance, we used a confusion matrix. This tells us that the model is **very strong at identifying non-cancelled rides**. It also does a good job predicting

cancelled rides with a few cancellations being missed (false negatives). Overall, the confusion matrix shows that the model has **high accuracy and very low error**, making it reliable for prediction.

We also analyzed **feature importance** to understand which factors influence cancellations the most. The results show:

- **Ride Distance** is the most important feature by a large margin
- **Avg CTAT (trip duration)** and **Avg VTAT (waiting time)** are also important
- Other features like **booking value, location, and payment method** have much smaller impact

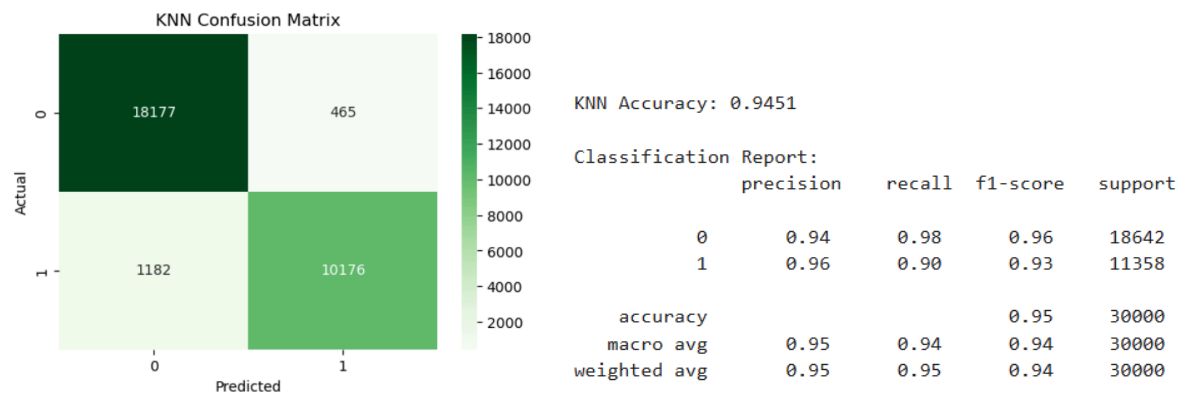
This means that:

- **Short rides are more likely to be cancelled**
- **Longer waiting times increase the chances of cancellation**
- Location and payment method matter, but not as much

KNN

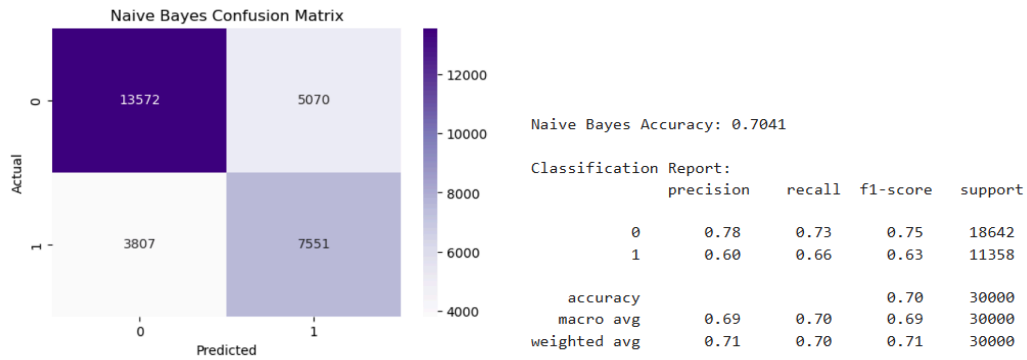
The KNN model predicts the chances of a ride cancellation. KNN works through identifying the nearest data points from the dataset and then predicts based on the average value of the neighbors. We cross validated the amount k needed in order to get the most accurate version of the model without overfitting. We got the model to have an overall accuracy of 95%. Notably it had a recall with 90% which meant it correctly identified 90% of cancelled rides. In the confusion matrix had 465 false positives indicating a good reliability for noncancellations. It is important to highlight there was a significant amount of false negatives so some cancellations

were missed, but the recall was still considerably high. Overall the model proved accurate in predicting. The drawback however was that in order for us to provide an accurate model, the feature importance portion of the code would not work. It would take too long to work due to the k value needed for the model to be accurate. There were ways around it by lowering the k values so it would take less time to run; however, it caused the model to be significantly less accurate.



Naive Bayes

Naive Bayes uses probability to predict the category of a data point. It will assume that the features are independent of one another. The model achieved a relatively accurate model with a percentage of 70.41%. The model had a relatively high amount of false positives (5070) and false negatives (3807). Due to the nature of the model assuming that the features are all independent of one another, it had considerably lower accuracy compared to other models. The lower precision and recall scores for class 1 indicates that the model had struggled to capture cancellation patterns.

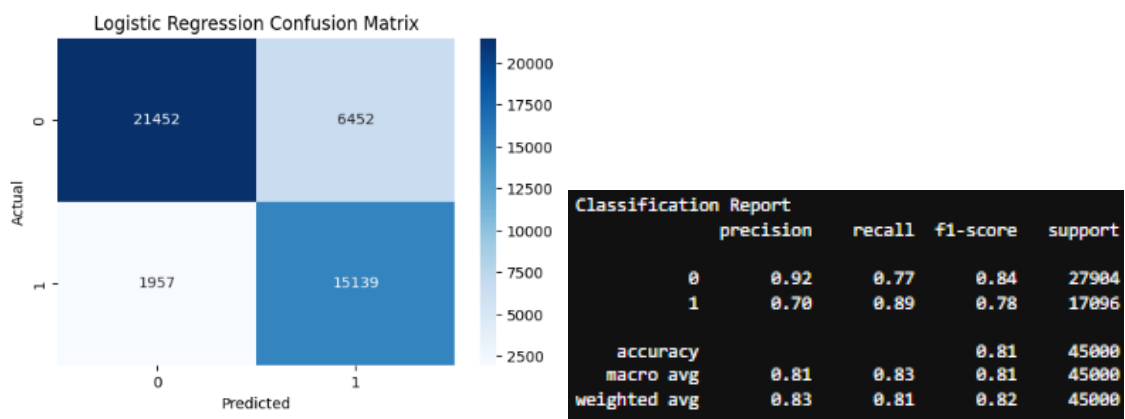


The feature importance for the Naive Bayes had originally shown that the features had significantly low impact on the data. So in order to fix this, the feature importance values were normalized in order to reflect the other model's values of the combined values of all features equating to 1. Notably, ride distance and Average CTAT were the top 2 most important features similar to other models. However, features like payment method, booking value, and drop location became more important than other features that were significant in other models like VTAT. Due to how Naive Bayes works, correlated variables like payment types and booking value may possibly dilute one another's influence. Some ways to improve on the model is with balance the classes through class weighting to reduce bias towards non-cancellations or for the model to switch to complement Naive Bayes since it is more robust for imbalance datasets to help improve recall.

Naive Bayes - Top 10 Normalized Feature Importance			
	feature	importance_raw	importance_normalized
0	Ride Distance	0.042447	0.325121
1	Avg CTAT	0.027523	0.210815
2	Payment Method_UPI	0.025257	0.193454
3	Booking Value	0.024533	0.187913
4	Payment Method_Uber Wallet	0.021680	0.166058
5	Payment Method_Credit Card	0.019533	0.149616
6	Payment Method_Debit Card	0.015300	0.117190
7	Pickup Location_Vinobapuri	0.000457	0.003498
8	Drop Location_Dwarka Mor	0.000413	0.003166
9	Pickup Location_Subhash Chowk	0.000357	0.002732

Logistic Regression

The Logistic Regression model predicts the probability of ride cancellation. The features were selected based on what information would be available at booking time, ensuring the model does not rely on post-outcome data and avoids data leakage. The target variable used was `is_cancelled`. We trained the model using a 70/30 train-test split and evaluated performance using a confusion matrix and classification report. The model achieved an overall accuracy of 81%. More importantly, it achieved a recall of 89% for cancellations, meaning it correctly identifies 89% of all cancelled rides. This is important because missing a cancellation is more costly than incorrectly predicting one. The model does produce more false positives than false negatives, but this trade-off is intentional, as it prioritizes capturing as many cancellations as possible. The model also performs well in identifying completed rides, with strong precision and recall. Overall, the model effectively predicts cancellations and can be useful for improving operational decision-making.



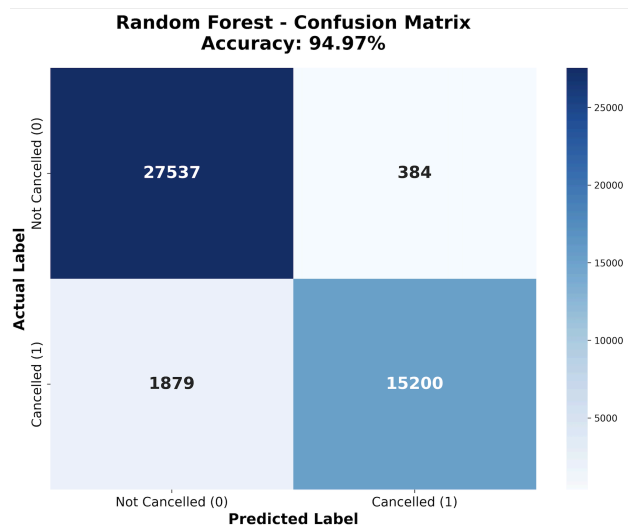
Using the logistic regression model, we identified several key factors that influence ride cancellations. The most important predictor was payment method, where digital payments are associated with significantly lower cancellation rates, suggesting that users are more committed

when payment is pre-set. We also found that shorter rides and lower booking values are more likely to be cancelled, while longer and higher-value rides show stronger commitment. Additionally, location and weekend patterns play a role, with certain areas and days showing higher cancellation rates. These features are associated with a lower/higher probability of cancellation. Overall, these insights help explain not just whether a ride will be cancelled, but why.

Top Features Increasing Cancellation (Positive Coefficients):			Top Features Decreasing Cancellation (Negative Coefficients):		
	Feature	Coefficient		Feature	Coefficient
364	Payment Method_UPI	1.412039	3	Ride Distance	-0.589528
5	day_of_week	0.035604	1	Avg CTAT	-0.559496
183	Pickup Location_Vinobapuri	0.033809	2	Booking Value	-0.335035
204	Drop Location_Bhikaji Cama Place	0.023193	0	Avg VTAT	-0.071794
135	Pickup Location_Pragati Maidan	0.022618	363	Payment Method_Debit Card	-0.025498
223	Drop Location_Ghaziabad	0.021211	91	Pickup Location_Khandsa	-0.025032
227	Drop Location_Govindpuri	0.020256	140	Pickup Location_RK Puram	-0.020510
210	Drop Location_Chhatarpur	0.017910	168	Pickup Location_Subhash Nagar	-0.018339
47	Pickup Location_GTB Nagar	0.017487	68	Pickup Location_IMT Manesar	-0.016487
35	Pickup Location_Chhatarpur	0.016738	295	Drop Location_Noida Extension	-0.016398

Random Forest

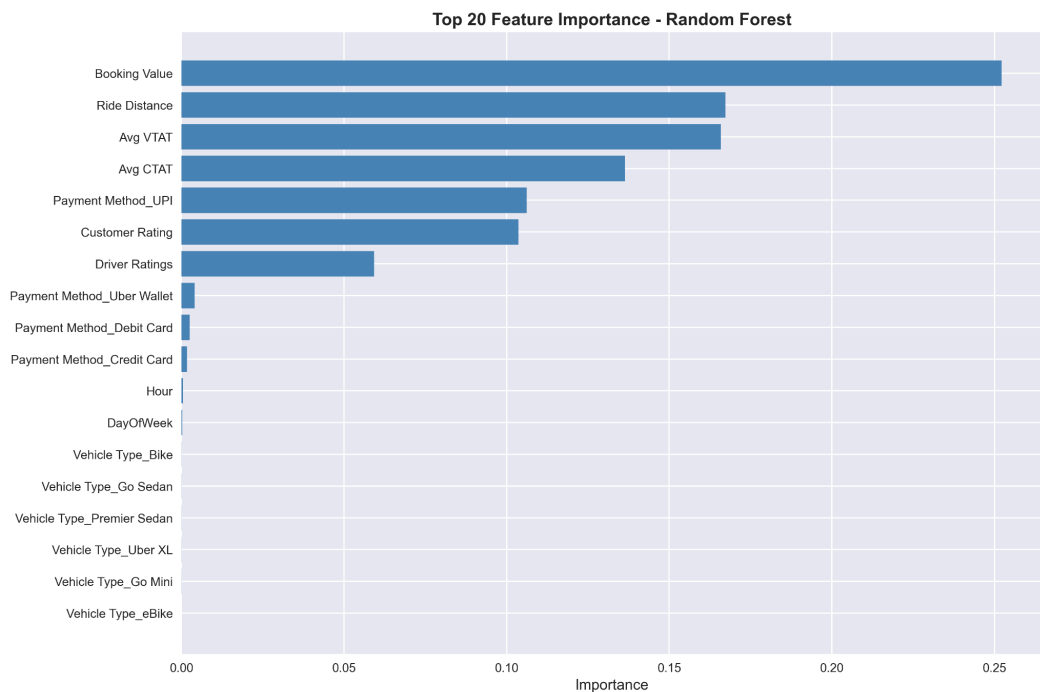
The Random Forest model was applied to predict whether a ride would be cancelled or not, and it achieved a testing accuracy of **94.97%**, indicating strong overall performance. This level of accuracy shows that the model is highly effective at capturing patterns in the data and generalizing them to unseen cases. The performance is especially notable given the complexity of user behavior involved in ride cancellations, which often depends on multiple interacting factors such as distance, time delays, and cost.



A closer look at the confusion matrix provides deeper insight into how the model performs across both classes. Out of all non-cancelled rides, **27,537 were correctly classified**, while

only **384 were incorrectly predicted as cancelled**, showing that the model is highly reliable in identifying rides that will not be cancelled. For cancelled rides, **15,200 were correctly identified**, while **1,879 were misclassified as non-cancelled**. This means the model performs very well overall, but it is slightly better at predicting non-cancellations than cancellations. In percentage terms, the model achieves very high precision for cancellations (about **97.5%**), meaning that when it predicts a cancellation, it is almost always correct. However, the recall is slightly lower (around **89%**), indicating that some actual cancellations are missed. This trade-off suggests the model is conservative when predicting cancellations, prioritizing correctness over capturing every possible case.

The feature importance results highlight which variables most strongly influenced the model’s predictions. **Ride Distance** is the most significant feature, contributing approximately **30.59%** of the total importance, making it the dominant factor in determining cancellation behavior. This suggests that trip length plays a major role in whether users follow through or cancel. The next most important feature is **Average CTAT**, contributing **24.59%**, which reflects the impact of time delays or estimated arrival times on user decisions. **Booking Value** follows with **20.49%**, indicating that cost is another critical factor influencing



cancellations. Together, these top three features account for over **75% of the model’s decision-making**, showing that the model relies heavily on a small set of highly informative variables.

Other features contribute significantly less. **Payment Method (UPI)** accounts for **13.11%**, suggesting some behavioral differences based on payment preference. **Average VTAT** contributes **2.95%**, while features like **Uber Wallet (2.79%)**, **Credit Card (2.35%)**, and **Debit Card (1.75%)** have relatively minor influence. Temporal features such as hour and day of the week each contribute less than **0.1%**, indicating that timing has very little impact compared to

Rank	Feature	Importance	Percentage
1	Ride Distance	0.305926	30.59%
2	Avg CTAT	0.245936	24.59%
3	Booking Value	0.204928	20.49%
4	Payment Method_UPI	0.131060	13.11%
5	Avg VTAT	0.029455	2.95%
6	Payment Method_Uber Wallet	0.027868	2.79%
7	Payment Method_Credit Card	0.023477	2.35%
8	Payment Method_Debit Card	0.017510	1.75%
9	hour	0.000993	0.10%
10	day_name_Sunday	0.000308	0.03%
11	day_name_Saturday	0.000264	0.03%
12	day_name_Monday	0.000237	0.02%
13	day_name_Tuesday	0.000210	0.02%
14	Vehicle Type_Go Mini	0.000209	0.02%
15	day_name_Wednesday	0.000208	0.02%

distance, delay, and cost. Similarly, vehicle type features contribute almost nothing, reinforcing that operational and pricing factors matter far more than ride category. The reason Random Forest achieved the highest accuracy lies in its ability to combine multiple

decision trees, each trained on different subsets of the data and features. This ensemble approach reduces overfitting and improves generalization, allowing the model to capture complex relationships without becoming too sensitive to noise. Unlike simpler models, Random Forest can model nonlinear interactions between key variables such as ride distance, waiting time, and booking value, which are critical in this problem. Additionally, the built-in randomness ensures that the model does not rely too heavily on any single feature, even though a few features dominate in importance.

Overall, the model demonstrates strong predictive power with a well-balanced performance across classes, though it slightly favors precision over recall for cancellations. The feature importance analysis confirms that user decisions are primarily driven by distance, time delays, and cost, while other variables play a minimal role. These results indicate that the Random Forest model is both effective and practical for predicting ride cancellations in a real-world setting.

Conclusion

Based on comparing the accuracies of the different models, we found that random forest was the most accurate. It did have a notable drawback being that the model took to run when compiling. So in exchange for more accuracy, the time took longer. This was highlighted more when compared to some of the other models like logistic regression and XGBoosting.

Throughout the models, we found the most prevalent features to be the ride distance, VTAT, and CTAT as the most important features that impacted Uber cancellations. This shows that what is most important for the customer and driver is the distance and time it takes for the ride.

Dataset Link:

<https://www.kaggle.com/datasets/yashdevladdha/uber-ride-analytics-dashboard?resource=download>